# Ecosystem

An ecosystem can be visualised as a functional unit of nature, where living organisms interact among themselves and also with the surrounding physical environment.

## How large, and how many kinds

- an ecosystem varies greatly in size, from a small pond to a large forest or a sea
- many ecologists regard the entire biosphere as a global ecosystem, a composite of all the local ecosystems on Earth
- since that system is too big and complex to be studied at one time, it is convenient to divide it into two basic categories, terrestrial and aquatic
- forest, grassland and desert are some terrestrial examples, pond, lake, wetland, river and estuary some aquatic ones, while crop fields and an aquarium may also be considered man-made ecosystems

## The order this chapter follows

- first the structure of the ecosystem, in order to appreciate the input, that is productivity, the transfer of energy, that is the food chain or web and nutrient cycling, and the output, that is degradation and energy loss
- then the relationships, the cycles, chains and webs, that are created as a result of these energy flows within the system, and their inter-relationship

## In this chapter

12.1 Structure of the ecosystem
12.3 Productivity
12.4 Decomposition
12.6 Energy flow
12.9 Trophic levels
12.11 Ecological pyramids

## 12.1 Structure of the ecosystem

### Physical structure

- the components of the environment are abiotic and biotic, and each affects the other and their surroundings
- the interaction of biotic and abiotic components results in a physical structure that is characteristic for each type of ecosystem
- identification and enumeration of the plant and animal species of an ecosystem gives its species composition
- the vertical distribution of different species occupying different levels is called stratification, so that trees occupy the top vertical strata of a forest, shrubs the second, and herbs and grasses the bottom layers

### The four aspects that make it a unit

- productivity
- decomposition
- energy flow
- nutrient cycling

## 12.2 The pond as an example

### A self-sustaining unit

- a small pond is fairly self-sustainable and a rather simple example that still explains the complex interactions of an aquatic ecosystem, being a shallow water body in which all four basic components are well exhibited
- the abiotic component is the water with all its dissolved inorganic and organic substances and the rich soil deposit at the bottom, while the solar input, the cycle of temperature, day-length and other climatic conditions regulate the rate of function of the entire pond
- the autotrophic components include the phytoplankton, some algae and the floating, submerged and marginal plants found at the edges, the consumers are the zooplankton and the free swimming and bottom dwelling forms, and the decomposers are the fungi, bacteria and flagellates especially abundant at the bottom

### What the pond performs

- conversion of inorganic into organic material with the help of the radiant energy of the sun by the autotrophs
- consumption of the autotrophs by heterotrophs
- decomposition and mineralisation of the dead matter, to release it back for reuse by the autotrophs, these events being repeated over and over again
- there is a unidirectional movement of energy towards the higher trophic levels and its dissipation and loss as heat to the environment

## 12.3 Productivity

### Primary production and productivity

- a constant input of solar energy is the basic requirement for any ecosystem to function and sustain
- primary production is the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis, expressed in terms of weight (g m–2) or energy (kcal m–2)
- the rate of biomass production is called productivity, expressed as gm–2 yr–1 or (kcal m–2) yr–1 so that different ecosystems can be compared

### GPP, NPP and secondary productivity

- gross primary productivity (GPP) of an ecosystem is the rate of production of organic matter during photosynthesis
- a considerable amount of GPP is utilised by plants in respiration, and gross primary productivity minus respiration losses (R) is the net primary productivity (NPP), that is GPP – R = NPP
- net primary productivity is the biomass available for consumption by heterotrophs, the herbiviores and decomposers
- secondary productivity is the rate of formation of new organic matter by consumers

### What productivity depends on

- primary productivity depends on the plant species inhabiting a particular area
- it also depends on a variety of environmental factors, on the availability of nutrients and on the photosynthetic capacity of plants, so it varies in different types of ecosystem
- the annual net primary productivity of the whole biosphere is approximately 170 billion tons of organic matter by dry weight
- of this, the oceans yield only 55 billion tons despite occupying about 70 per cent of the surface, and the rest is on land, the main reason for the low productivity of the ocean being a matter to discuss with your teacher

## 12.4 Decomposition

### Detritus, and the five steps

- the earthworm is referred to as the farmer's friend because such animals help break down complex organic matter and loosen the soil
- decomposers likewise break down complex organic matter into inorganic substances such as carbon dioxide, water and nutrients, and the process is called decomposition
- dead plant remains such as leaves, bark and flowers, and the dead remains of animals including fecal matter, constitute detritus, the raw material for decomposition
- the important steps are fragmentation, leaching, catabolism, humification and mineralisation

### The first three steps

- detritivores such as the earthworm break detritus down into smaller particles, which is fragmentation
- by leaching, water-soluble inorganic nutrients go down into the soil horizon and are precipitated as unavailable salts
- bacterial and fungal enzymes degrade detritus into simpler inorganic substances, which is catabolism
- all of these steps operate simultaneously on the detritus, while humification and mineralisation occur during decomposition in the soil

**Detritus**, the raw material entering the cycle

**Fragmentation and leaching**, acting on it in the soil

**Catabolism by bacterial and fungal enzymes**

**Humification**, building the dark humus

**Mineralisation**, releasing inorganic nutrients back

**Figure 12.A** Diagrammatic representation of decomposition cycle in a terrestrial ecosystem

## 12.5 Humus and mineralisation

### Humus

- humification leads to the accumulation of a dark coloured amorphous substance called humus
- humus is highly resistant to microbial action and undergoes decomposition at an extremely slow rate
- being colloidal in nature it serves as a reservoir of nutrients
- humus is further degraded by some microbes, and the release of inorganic nutrients then occurs by the process known as mineralisation

### What controls the rate

- decomposition is largely an oxygen-requiring process
- the rate is controlled by the chemical composition of the detritus and by climatic factors
- in a particular climatic condition the rate is slower if the detritus is rich in lignin and chitin, and quicker if it is rich in nitrogen and water-soluble substances such as sugars
- temperature and soil moisture are the most important climatic factors, acting through their effects on the activities of soil microbes, so a warm and moist environment favours decomposition while low temperature and anaerobiosis inhibit it and organic materials build up

## 12.6 Energy flow

### The sun, and how little of it is used

- except for the deep sea hydro-thermal ecosystem, the sun is the only source of energy for all ecosystems on Earth
- less than 50 per cent of the incident solar radiation is photosynthetically active radiation (PAR)
- plants and photosynthetic bacteria, the autotrophs, fix the sun's radiant energy to make food from simple inorganic materials, capturing only 2-10 per cent of the PAR, and this small amount of energy sustains the entire living world
- all organisms depend for their food on producers, directly or indirectly, so the flow of energy from the sun to producers and then to consumers is unidirectional, which is worth weighing against the first law of thermodynamics

### The second law as well

- ecosystems are not exempt from the second law of thermodynamics
- they need a constant supply of energy to synthesise the molecules they require
- that supply counteracts the universal tendency toward increasing disorderliness

## 12.7 Producers and consumers

### Producers, and the chain that starts with them

- the green plants of an ecosystem are called producers, the major ones in a terrestrial ecosystem being herbaceous and woody plants, and in an aquatic ecosystem species such as phytoplankton, algae and higher plants
- starting from the producers, food chains or rather webs are formed such that an animal feeds on a plant or on another animal and in turn is food for another, and the chain or web is formed because of this interdependency
- no energy trapped into an organism remains in it for ever, so the energy trapped by the producer is either passed on to a consumer or the organism dies, and the death of an organism is the beginning of the detritus food chain or web

### The ranks of consumer

- all animals depend on plants, directly or indirectly, for their food needs, and are hence called consumers, and also heterotrophs
- those that feed on the producers are primary consumers, those that eat other animals which in turn eat the plants are secondary consumers, and there can be tertiary consumers too
- the primary consumers are the herbivores, some common ones being insects, birds and mammals in terrestrial ecosystems and molluscs in aquatic ones
- the consumers that feed on these herbivores are carnivores, or more correctly primary carnivores though secondary consumers, and those that depend on the primary carnivores are secondary carnivores, so a simple grazing food chain (GFC) runs grass, the producer, to goat, the primary consumer, to man, the secondary consumer

**First trophic level**, the producers

**Second trophic level**, the herbivores or primary consumers

**Third trophic level**, the carnivores or secondary consumers

**Energy passing upward**, less of it at each step

**Figure 12.B** Diagrammatic representation of trophic levels in an ecosystem

## 12.8 The detritus food chain and the food web

### The detritus food chain

- the detritus food chain (DFC) begins with dead organic matter
- it is made up of decomposers, heterotrophic organisms, mainly fungi and bacteria, which meet their energy and nutrient requirements by degrading dead organic matter or detritus, and are also known as saprotrophs, from sapro, to decompose
- decomposers secrete digestive enzymes that break down dead and waste materials into simple inorganic materials, which they then absorb

### How the chains join into a web

- in an aquatic ecosystem the GFC is the major conduit for energy flow, whereas in a terrestrial ecosystem a much larger fraction of energy flows through the detritus food chain than through the GFC
- the detritus food chain may be connected with the grazing food chain at some levels, since some organisms of the DFC are prey to GFC animals
- in a natural ecosystem some animals such as cockroaches and crows are omnivores, and these natural interconnections of food chains make it a food web, which raises the question of where human beings belong in such a classification

## 12.9 Trophic levels

### A place set by feeding

- organisms occupy a place in the natural surroundings, or in a community, according to their feeding relationship with other organisms
- based on the source of their nutrition or food, an organism occupies a specific place in the food chain, known as its trophic level
- producers belong to the first trophic level, herbivores, the primary consumers, to the second, and carnivores, the secondary consumers, to the third

**Producer at the base**, holding the most energy

**Herbivore next**, receiving a tenth of it

**Primary carnivore above that**

**Secondary carnivore at the top of the grazing food chain**

**Figure 12.C** Energy flow through different trophic levels

## 12.10 Standing crop and the ten per cent law

### Energy down the levels

- the amount of energy decreases at successive trophic levels
- when any organism dies it is converted to detritus or dead biomass, which serves as an energy source for decomposers
- organisms at each trophic level depend on those at the lower trophic level for their energy demands

### Standing crop

- each trophic level has a certain mass of living material at a particular time, called the standing crop
- the standing crop is measured as the mass of living organisms, that is biomass, or as the number in a unit area
- the biomass of a species is expressed in terms of fresh or dry weight, and measurement in terms of dry weight is more accurate

### The ten per cent law

- the number of trophic levels in the grazing food chain is restricted, because the transfer of energy follows the 10 per cent law
- only 10 per cent of the energy is transferred to each trophic level from the level below it
- in nature it is possible to have producer, herbivore, primary carnivore and secondary carnivore in the grazing food chain, and whether the detritus food chain carries any such limitation is worth thinking through

## 12.11 Ecological pyramids

### The shape

- the base of a pyramid is broad and it narrows towards the apex
- one gets a similar shape whether the food or the energy relationship between organisms at different trophic levels is expressed

**Grassland producers**, nearly 6 millions plants at the base

**Herbivores above them**, far fewer

**Carnivores above those**

**Only three top-carnivores at the apex**

**Figure 12.D** Pyramid of numbers in a grassland ecosystem. Only three top-carnivores are supported in an ecosystem based on production of nearly 6 millions plants

**Producer biomass**, the broad base

**Herbivore biomass**, sharply less

**Carnivore biomass**, less again

**Figure 12.E** Pyramid of biomass shows a sharp decrease in biomass at higher trophic levels

**Phytoplankton**, a small standing crop at the base

**Zooplankton**, a large standing crop above it

**The figure standing on its apex**, hence inverted

**Figure 12.F** Inverted pyramid of biomass-small standing crop of phytoplankton supports large standing crop of zooplankton

## 12.12 The three pyramids

### What each pyramid expresses

- the relationship is expressed in terms of number, biomass or energy
- the base of each pyramid represents the producers, that is the first trophic level, while the apex represents the tertiary or top level consumer
- the three types usually studied are the pyramid of number, the pyramid of biomass and the pyramid of energy

**Sunlight arriving at the producers**

**Primary producers**, converting only 1% of it into NPP

**Each higher bar smaller than the one below**

**Energy lost as heat at every step**

**Figure 12.G** An ideal pyramid of energy. Observe that primary producers convert only 1% of the energy in the sunlight available to them into NPP

## 12.13 Reading the pyramids

### Count everything at the level

- any calculation of energy content, biomass or numbers has to include all the organisms at that trophic level
- no generalisation we make will be true if only a few individuals at any trophic level are taken into account
- a given organism may occupy more than one trophic level at the same time, since a trophic level represents a functional level and not a species as such
- a sparrow, for instance, is a primary consumer when it eats seeds, fruits and peas and a secondary consumer when it eats insects and worms, which raises the question of how many trophic levels human beings function at

### Upright, and the exceptions

- in most ecosystems all the pyramids, of number, of energy and of biomass, are upright, so producers are more in number and biomass than the herbivores, and herbivores more than the carnivores, and energy at a lower trophic level is always more than at a higher one
- there are exceptions to this generalisation, as counting the insects feeding on a big tree, then the small birds depending on those insects and the larger birds eating the smaller ones will show
- the pyramid of biomass in the sea is generally inverted, because the biomass of fishes far exceeds that of phytoplankton, which looks like a paradox and calls for an explanation

### The pyramid of energy

- the pyramid of energy is always upright and can never be inverted
- this is because when energy flows from a particular trophic level to the next, some energy is always lost as heat at each step
- each bar in the energy pyramid indicates the amount of energy present at each trophic level in a given time, or annually per unit area

## 12.14 Limitations of ecological pyramids

### What a pyramid leaves out

- it does not take into account the same species belonging to two or more trophic levels
- it assumes a simple food chain, something that almost never exists in nature, and it does not accommodate a food web
- saprophytes are given no place in ecological pyramids, even though they play a vital role in the ecosystem

## 12.15 Summary

### Structure

- an ecosystem is a structural and functional unit of nature, comprising abiotic and biotic components, the abiotic being inorganic materials, that is air, water and soil, and the biotic being producers, consumers and decomposers
- each ecosystem has a characteristic physical structure resulting from the interaction among abiotic and biotic components, and species composition and stratification are its two main structural features
- based on the source of nutrition, every organism occupies a place in an ecosystem, and productivity, decomposition, energy flow and nutrient cycling are the four important components of one

### Function

- primary productivity is the rate of capture of solar energy or biomass production of the producers, divided into gross primary productivity (GPP), the rate of capture of solar energy or total production of organic matter, and net primary productivity (NPP), the biomass or energy left after utilisation by the producers, while secondary productivity is the rate of assimilation of food energy by the consumers
- in decomposition the complex organic compounds of detritus are converted to carbon dioxide, water and inorganic nutrients by the decomposers, and it involves three processes, fragmentation of detritus, leaching and catabolism
- energy flow is unidirectional, plants first capturing solar energy and food then being transferred from the producers to the decomposers, with organisms of different trophic levels connected to each other for food or energy in a food chain
- the storage and movement of nutrient elements through the various components of the ecosystem is nutrient cycling, of two types, gaseous and sedimentary, the atmosphere or hydrosphere being the reservoir for the gaseous type such as carbon and the Earth's crust for the sedimentary type such as phosphorus, while the products of ecosystem processes are named ecosystem services, for example the purification of air and water by forests

## 12.16 Exercises

### From this chapter

1 Fill in the blanks. (a) Plants are called as_____because they fix carbon dioxide. (b) In an ecosystem dominated by trees, the pyramid (of numbers) is_____type.

## 12.17 Exercises, continued

### From this chapter, continued

1 (c) In aquatic ecosystems, the limiting factor for the productivity is_____. (d) Common detritivores in our ecosystem are_____. (e) The major reservoir of carbon on earth is_____.
2 Which one of the following has the largest population in a food chain? (a) Producers (b) Primary consumers (c) Secondary consumers (d) Decomposers
3 The second trophic level in a lake is (a) Phytoplankton (b) Zooplankton (c) Benthos (d) Fishes
4 Secondary producers are (a) Herbivores (b) Producers (c) Carnivores (d) None of the above
5 What is the percentage of photosynthetically active radiation (PAR) in the incident solar radiation? (a) 100% (b) 50 % (c) 1-5% (d) 2-10%
6 Distinguish between (a) Grazing food chain and detritus food chain (b) Production and decomposition (c) Upright and inverted pyramid (d) Food chain and Food web (e) Litter and detritus (f) Primary and secondary productivity
7 Describe the components of an ecosystem.
8 Define ecological pyramids and describe with examples, pyramids of number and biomass.
9 What is primary productivity? Give brief description of factors that affect primary productivity.
10 Define decomposition and describe the processes and products of decomposition.
11 Give an account of energy flow in an ecosystem.